Asian College of Higher Studies

Ekantakuna, Lalitpur

Lab Report

Department : Computer Science & Information Technology

Course Title : Software Project Management

Course No : CSC 425

Experiment No : 03

Experiment Name: Project Scheduling : Gantt Chart & Network Diagram

Submitted By	Submitted To
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Level: Bachelor	*Signature:* ....
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Date of Submission: March 3	

Lab 3 - Project Scheduling : Gantt Charts and Network Diagram

Q. 1) Draw a Gantt chart & network diagram for the activities with given relationships as shown in the following table:

Activity	Duration (weeks)	Predecessor
A	5	—
B	8	A
C	5	B
D	10	B
E	8	C, D
F	5	E

Solution:

Schedule Calculation:

Activity	ES Calculation	ES	EF
A	No predecessor → ES = 0	0	5
B	After A → ES = EF(A) = 5	5	13
C	After B → ES = EF(B) = 13	13	18
D	After B → ES = EF(B) = 13	13	23
E	max{EF(C) = 18, EF(D) = 23} = 23	23	31
F	After E → ES = EF(E) = 31	31	36

∴ Critical path = A → B → D → E - F = 5 + 8 + 10 + 8 + 5
= 36 weeks

Gantt Chart:

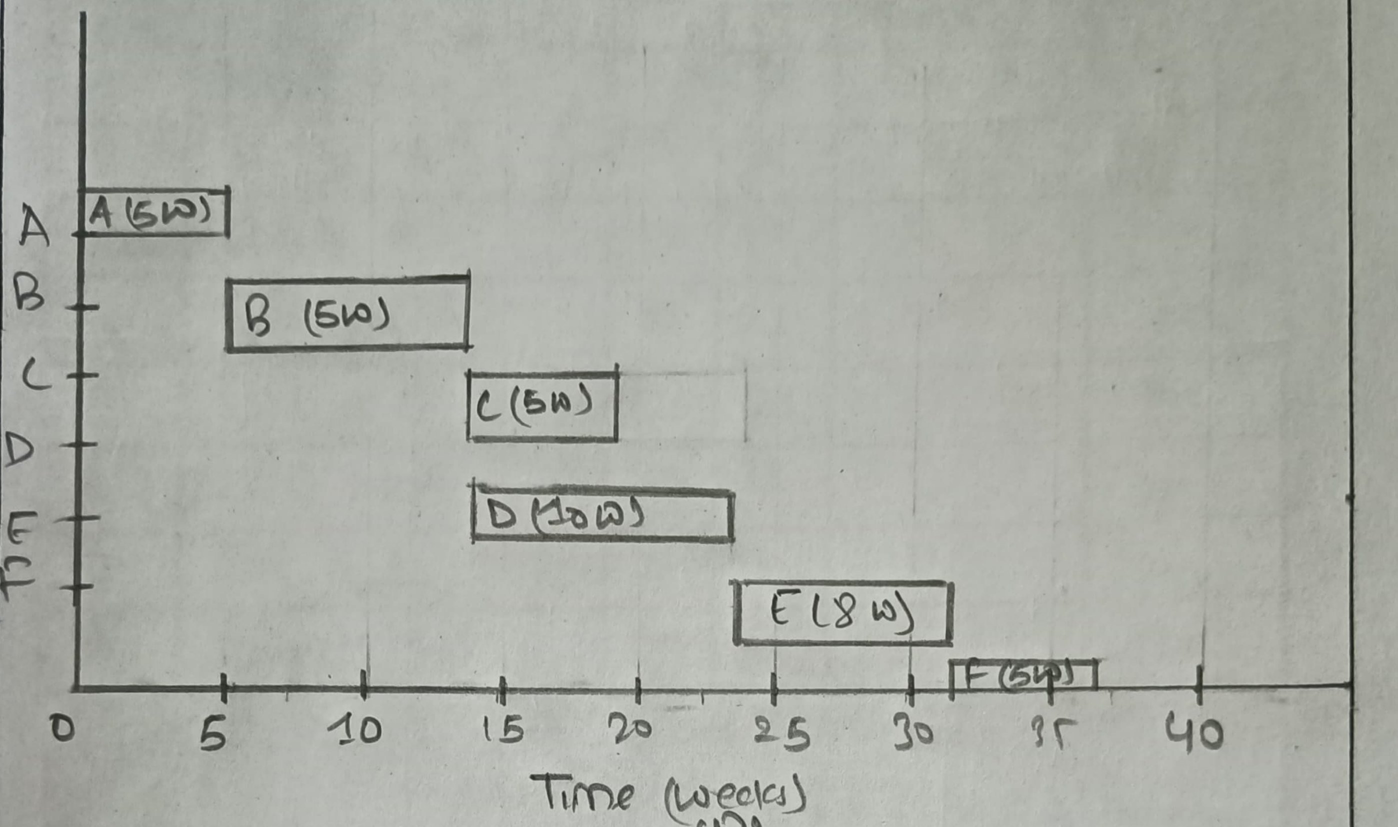
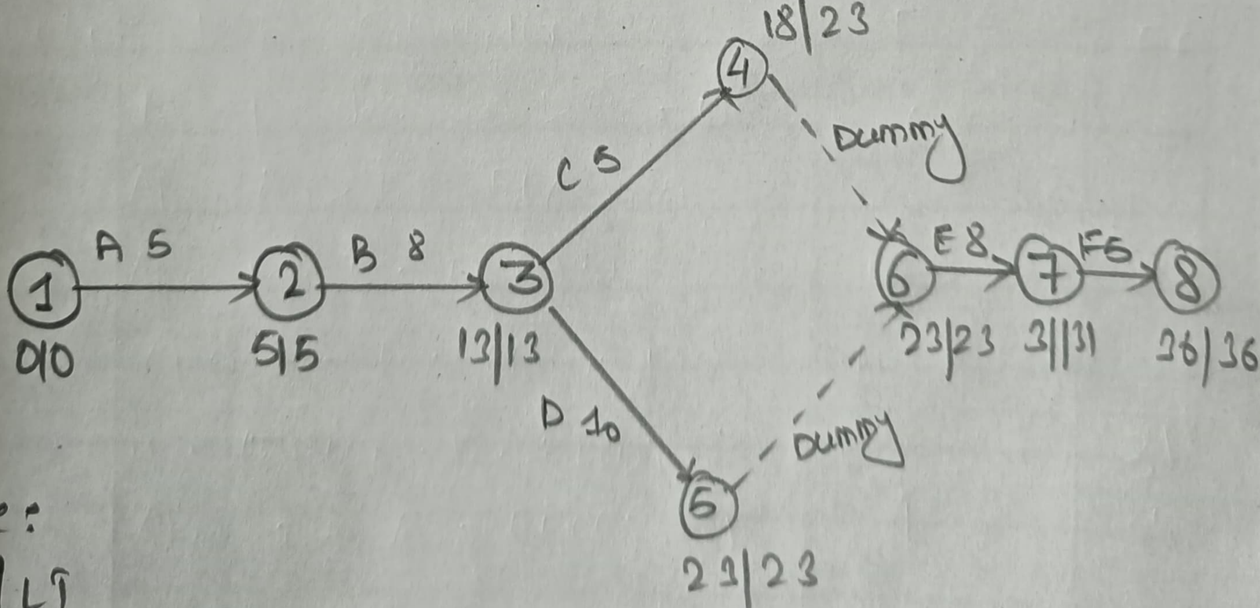


Fig: Gantt Chart Q1

Network Diagram:



Note:
ET/LT

Fig: Network Diagram Q1

Node	Represents	ET	LT	Float = LT-ET	Critical
1	Project Start	0	0	0	Yes
2	End of A	5	5	0	Yes
3	End of B	13	13	0	Yes
4	End of C	18	23	005	No
5	End of D	23	23	0	Yes
6	C & D Merge	23	23	0	Yes
7	End of F	31	31	0	Yes
8	Project End	36	36	0	Yes

∴ Critical Path: $1 \to 2 \to 3 \to 5 \to 6 \to 7 \to 8 =$

$A + B + D + F = 5 + 8 + 10 + 8 + 5$

$= 36$ weeks

Q.2) Draw a gantt chart & network diagram for the activities with given relationship as shown in the following table.

Activity	Predecessor	Duration (Weeks)
A	–	4
B	–	5
C	A	5
D	A	6
E	B, C	5
F	D	4
G	E	5

Solution:

Schedule Calculation:

Activity	ES Calculation	ES	EF
A	No predecessor → ES → 0	0	4
B	No predecessor → ES = 0	0	5
C	After A → ES = EF(A) = 4	4	9
D	After A → ES = EF(A) = 4	4	10
E	max(EF(B) = 5, EF(C) = 9) = 9	9	14
F	After D → ES = EF(D) = 10	10	14
G	After E → ES = EF(E) = 14	14	19

∴ Critical Path → A → D → F → G = 4 + 6 + 4 + 5 = 19 weeks

Gantt chart:

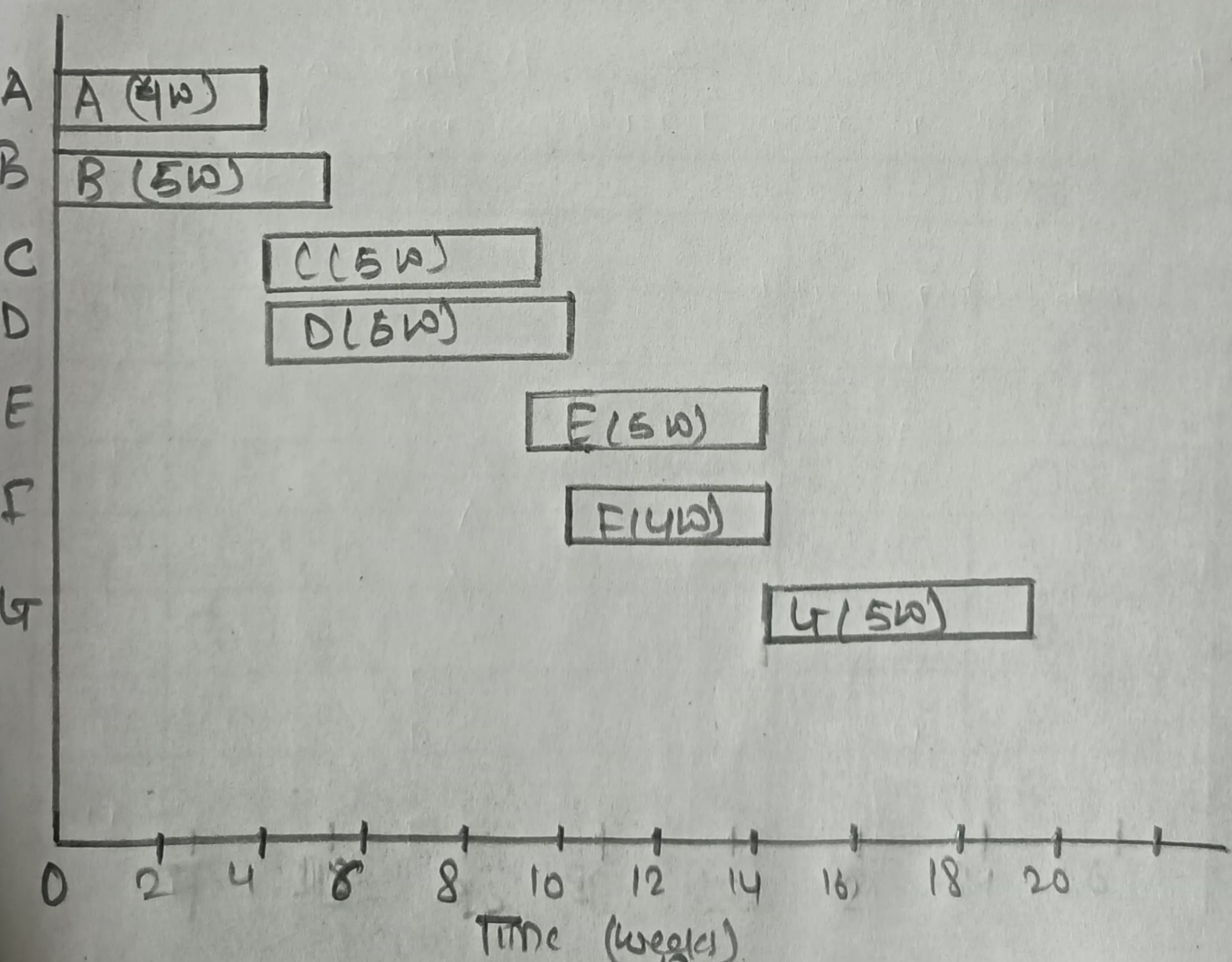


Fig: Gantt Chart Q2

Network Diagram:

Note:
ET|LT

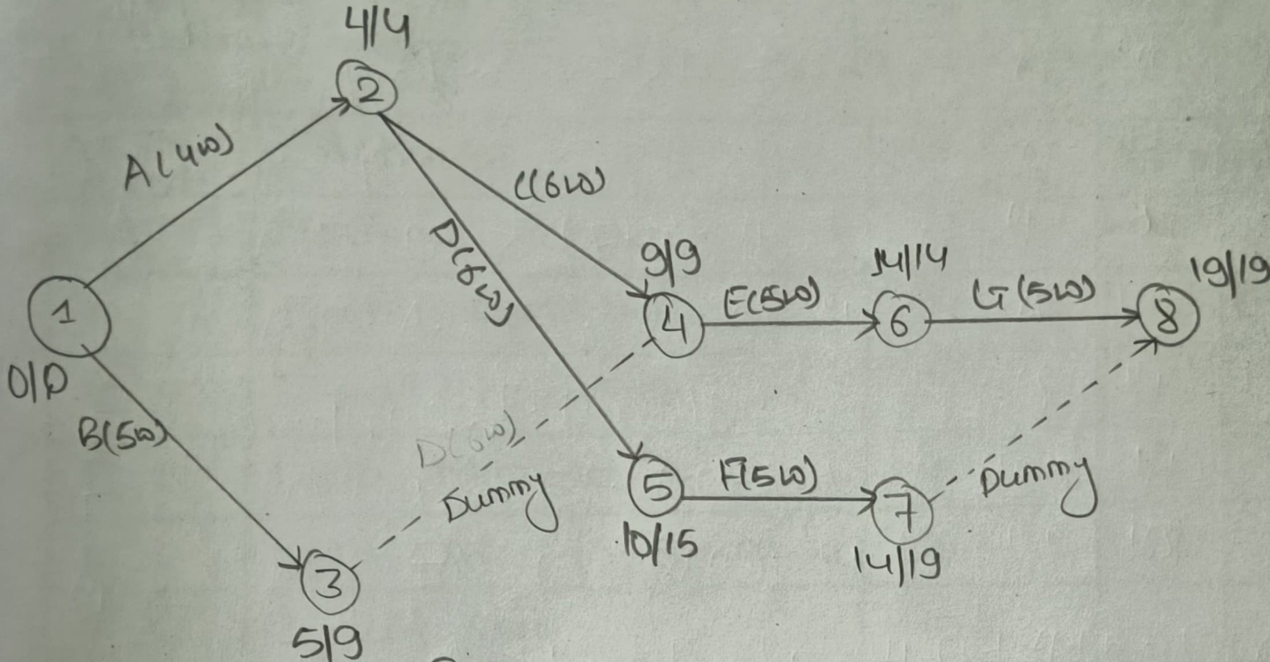


Fig: Network Diagram Q2

Critical Path : 1->2->4->6->8 | A+C+E+G = 4+6+5+5
= 19 weeks

Node	Represents	ET	LT	Float->LT-ET	Critical
1	Project Start	0	0	0	Yes
2	End of A	4	4	0	Yes
3	End of B	5	9	4	No
4	B+C merge - E can start	9	9	0	Yes
5	End of D	10	15	5	No
6	End of E	14	14	0	Yes
7	End of F	14	19	5	No
8	Project End	19	19	0	Yes

ES, EF → Forward Pass, LS/LF → Backward

Q3) The table below is an example of project specification with estimated activity duration & precedence requirements. Draw Gantt Chart & precedence network diagram & identify the critical path of the project & calculate the earliest completion time of the project.

Activity Name	Activity	Duration (Weeks)	Precedents
A	Hardware Selection	6	
B	System Configuration	4	
C	Install Hardware	3	
D	Data Migration	4	A
E	Draft office Procedures	3	B
F	Recruit staff	10	
G	User Training	3	E, F
H	Install & test system	2	C, D

<u>Solution:</u>

Schedule Calculation:

Activity	Duration	ES	EF	LS	LF	Float (LF-EF)	Critical
A	6	0	6	1	7	1	No
B	4	0	4	3	7	3	No
C	3	0	3	8	11	8	No
D	4	6	10	7	11	1	No
E	3	4	7	7	10	3	No
F	10	0	10	0	10	0	~~No~~ Yes
G	3	10	13	10	13	0	Yes
H	2	10	12	11	13	1	No

Gantt Chart:

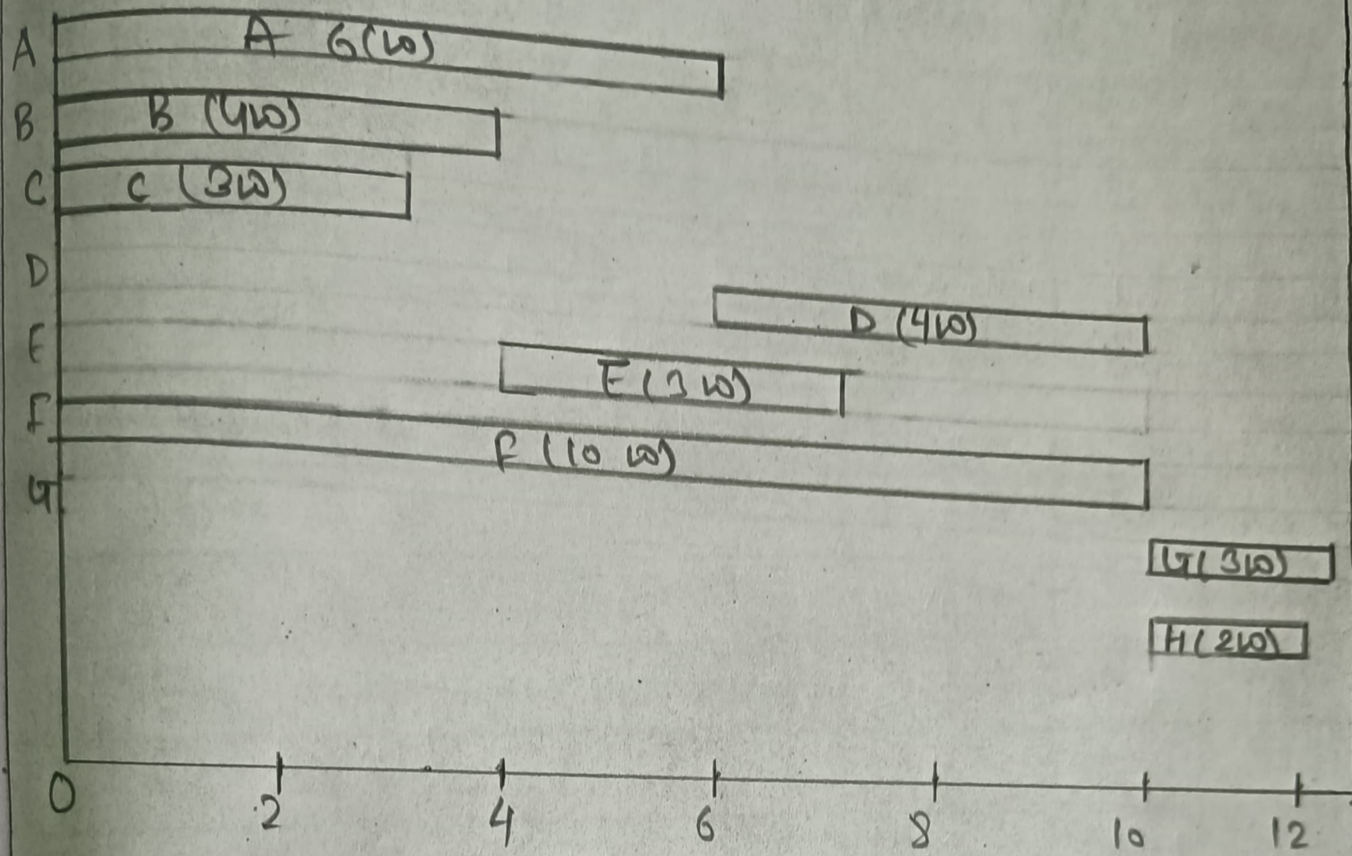


Fig: Gantt Chart Q 3

Network Diagram:

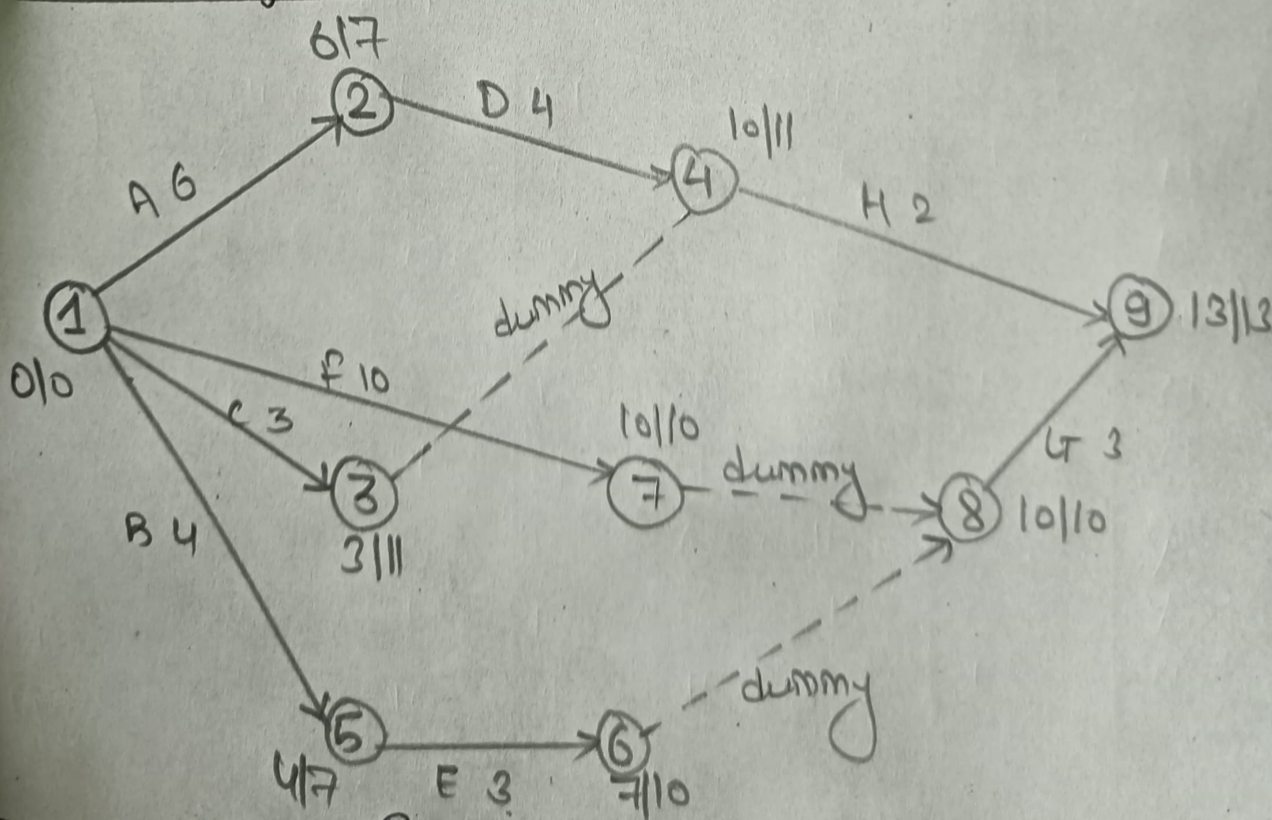


Fig: Network Diagram Q3

∴ Critical Path: 1 -> 7 -> 8 -> 9 | 10+0+3 = 13 weeks

Node	Represents	ET	LT	Float = LT-ET	Critical
1	Project Start	0	0	0	Yes
2	End of A	6	7	1	No
3	End of C	3	11	8	No
4	C+D merge -> H Can start	10	11	1	No
5	End of B	4	7	3	No
6	End of E	7	10	3	No
8	E+F merge -> G can start	10	10	0	Yes
7	End of F	10	10	0	Yes
9	Project End	13	13	0	Yes